Assignment

① Write a brief on Primary, foreign, candidate and Super key.

① **Primary key** → A primary key is a column or a combination of coloumns that uniquely identifies each row in a table.

* It does not allow duplicate values or Null values.
* Just as every person has a unique aadhar number, every row in a table has a Primary key that value is unique.

Characteristics
* Every primary key value must be different.
* Every row must have a primary key value.
* Every row must have only one primary key.
* A table can have only one primary key can be:
  - that one primary key
    - One Coloumn (Single-column Primary key)
    - Or Multiple Coloumn together (Composite primary key).

Ex→ Primary key (Student_id).
Primary key (student_id, course_id).

② Foreign keys. * Its main purpose is to create a relationship between two tables.

* A foreign key is a Column or a Combination of Columns in one table that refers to the primary key (or a unique) key of another table.

⟹ A foreign key is a field in a child table that references the Primary key of a Parent table.

* Parent table contains the Primary key.
* Child table contains the foreign key.

Rules of a foreign key

① Reference a Primary key.
② Duplicate values are allowed.
③. Null values are allowed.
④ A table can have multiple foreign keys.

③ Candidate key → A candidate key is the minimum set of columns that can uniquely identities each row in a table.

* A table can have multiple candidate keys, but only one of them is selected as the primary key.

minimum set of column needed for uniqueness.

Important Points

* It must uniquely identify every row.
* It can contain extra columns that are not required for uniqueness.

Ex. Student_id	Email	Name	Age
101	riya@g	riya	20
102	Aman@gm	Aman	21
103	Priya@m	Priya	22

Super keys - Student_id
Email
Student id + Name
Student_id + Age
Student_id + Name
Email + Age
Email + Name
Student_id + Name + Email.

Characteristics

1. Must uniquely identify every row.
2. Can consist of one or more columns.
3. May contain Extra unnecessary columns.
4. A table can have many Super keys.

Super key
A column or combination of columns that

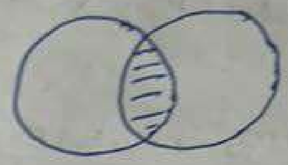
Candidate key

	Primary key.	foreign key.	Candidate key.	Super key.
Definition	A key Selected to uniquely identify each row in a table.	A key in one table that refers to the Primary key of another table.	Minimum set of Columns that uniquely identify every row in a table.	A column or combination of columns that or unnecessary columns that uniquely identify rows.
Purpose	Uniquely identifies every record in a table	Create a relationship between two tables.	Provide all possible choices for the Primary key.	identifies rows uniquely - contains unnecessary columns.
** Uniqueness	Must be unique.	Duplicate values are allowed	Must be unique.	Must be unique.
Null values.	Not allowed	Allowed	Not allowed	Depends on the column used.
Duplicate values	Not allowed	Allowed	Not allowed.	Not allowed.
No. in a table	Only one Primary key.	multiple.	multiple Candidate key Subset of Super keys.	multiple.

(5) Combine data from multiple tables.
(6) Save storage space.
(7) Improve database design.

Types of Joins In SQL

(1) <u>Inner Join</u> - Returns only the rows that have a match in both tables. If there is no match, SQL ignores that row.



(2) <u>Left Join</u> (Left outer join) → All rows from the left table.

+ Only the matching row from the right table.

+ If there is no matching row in the right table, SQL returns <u>null</u> for the right table's column.

(3) Right Join (Right Outer Join) →

+ All rows from the right table.

+ Only the matching rows from the left table.

Null value appears in	Right table Columns	Left table Columns
Main Purpose	Keep all records from the left table	Keep all records from the right table.

Q. What is Self Join? → A Self join is a type of SQL join in which a table is joined with itself.

Unlike inner Join, left join, or Right join, which join two different tables, a Self join treats one table as if it were two Separate tables by using table aliases.

**. A self join is a join where a table is joined to itself using aliases to compare or relate rows with in the Same table.

Why do we need a Self Join → Sometimes, all the required information is stored in one table, but one row needs to be compared with another row in the same table.

Employees table

Emp_id	Employee_Name	Manager_id
1	John	Null
2	Alice	1
3	Bob	1
4	Charlie	2
5	David	2
6	Emma	3